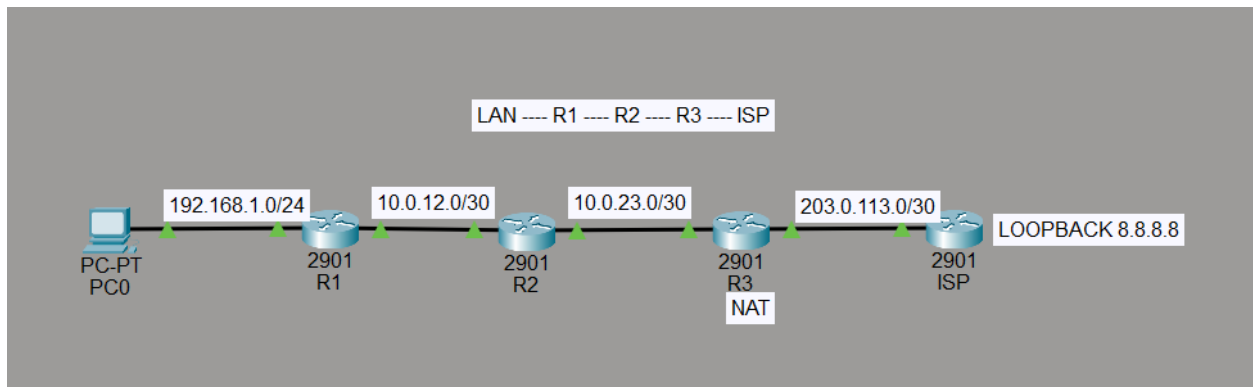


MINI PROJECT 12

OSPF(Single area), INJECT_STATIC_DEFAULT_WITH_PAT

DIAGRAM:



DESCRIPTION:

It is a small enterprise network where internal routers (R1–R2–R3) ran OSPF to share internal routes dynamically, while R3 acted as the edge router connected to an ISP. On R3, we configured a static default route toward the ISP and injected that default into OSPF so internal routers could reach the internet. We also implemented NAT (PAT) on R3 to translate private LAN addresses into a public IP which is almost similar to real world network setup.

STEP 1:Configure interfaces for R1, R2, R3 & ISP:

R1:

```
Int g0/0
Ip address 192.168.1.1 255.255.255.0
No shut
```

```
Int g0/1
Ip address 10.0.12.1 255.255.255.252
No shut
```

R2:

```
Int g0/1  
Ip address 10.0.12.2 255.255.255.252  
No shut
```

```
Int g0/0  
Ip address 10.0.23.1 255.255.255.252  
No shut
```

R3:

```
Int g0/0  
Ip address 10.0.23.2 255.255.255.252  
Ip nat inside  
No shut
```

```
Int g0/1  
Ip address 203.0.113.1 255.255.255.252  
Ip nat outside  
No shut
```

ISP:

```
Int g0/1  
Ip address 203.0.113.1 255.255.255.252  
No shut
```

```
Int loopback 0  
Ip address 8.8.8.8 255.255.255.255
```

STEP 2: Configure OSPF for R1, R2, R3:

R1:

```
router ospf 1  
network 192.168.1.0 0.0.0.255 area 0  
network 10.0.12.0 0.0.0.3 area 0
```

R2:

```
router ospf 1
network 10.0.12.0 0.0.0.3 area 0
network 10.0.23.0 0.0.0.3 area 0
```

R3:

```
router ospf 1
network 10.0.23.0 0.0.0.3 area 0
```

Note: Don't advertise ISP Interface.

STEP 3: Configure Static Default on R3:

```
ip route 0.0.0.0 0.0.0.0 203.0.113.2
```

Inject default into OSPF: Other routers can learn the static route

```
router ospf 1
default-information originate
```

STEP 4: Configure PAT on R3:

```
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface g0/1 overload
```

Note: With NAT, we don't have to configure routers with static routes

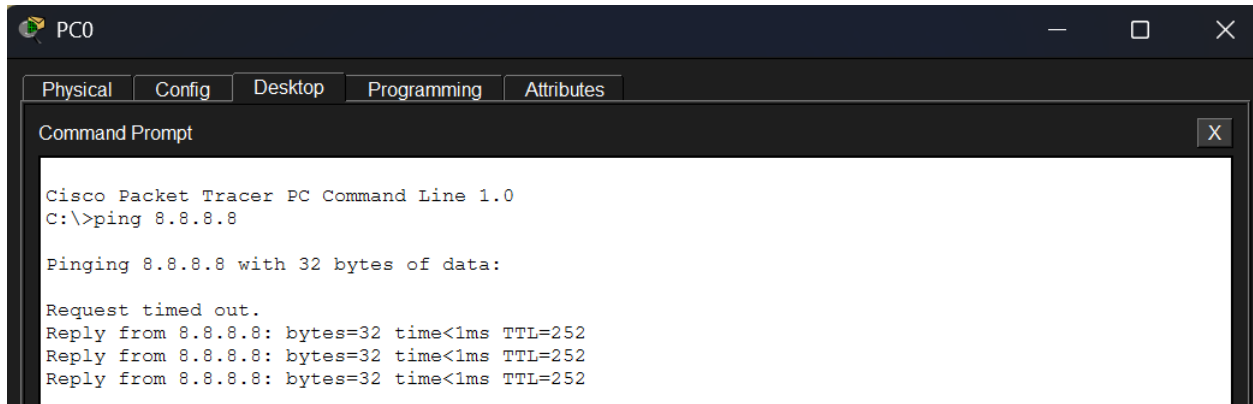
STEP 5 : Client device configure:

PC	VLAN	IP Address	Gateway
PC0	-	192.168.1.10/24	192.168.1.1

STEP 6: Verification & ping test:

```
Show ip route
Show ip ospf neighbor
Show ip nat translation (on R3)
```

Ping test from PC0:



The screenshot shows a PC0 window with tabs for Physical, Config, Desktop, Programming, and Attributes. The Command Prompt is open, displaying the output of a ping command to 8.8.8.8. The output indicates that the ping was successful with a response time of less than 1ms and a TTL of 252.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Request timed out.
Reply from 8.8.8.8: bytes=32 time<1ms TTL=252
Reply from 8.8.8.8: bytes=32 time<1ms TTL=252
Reply from 8.8.8.8: bytes=32 time<1ms TTL=252
```

Nat translation:

```
Router#sh ip nat translations
Pro  Inside global      Inside local      Outside local     Outside global
icmp 203.0.113.1:1      192.168.1.10:1   8.8.8.8:1        8.8.8.8:1
icmp 203.0.113.1:2      192.168.1.10:2   8.8.8.8:2        8.8.8.8:2
icmp 203.0.113.1:3      192.168.1.10:3   8.8.8.8:3        8.8.8.8:3
icmp 203.0.113.1:4      192.168.1.10:4   8.8.8.8:4        8.8.8.8:4
Router#
```

Note: This mini enterprise project is focused on OSPF static inject with NAT and not exact real world network topology.